

Production Supporting Systems in Factories

ระบบสนับสนุนการผลิตในโรงงานอุตสาหกรรม

Machine Learning

| Image Classification

Setting up ML server

- Get [code](#)
- `pnpm install`
- `pnpm run build`
- `pnpm run start`
- Test if server is running.
 - <http://localhost:3003>

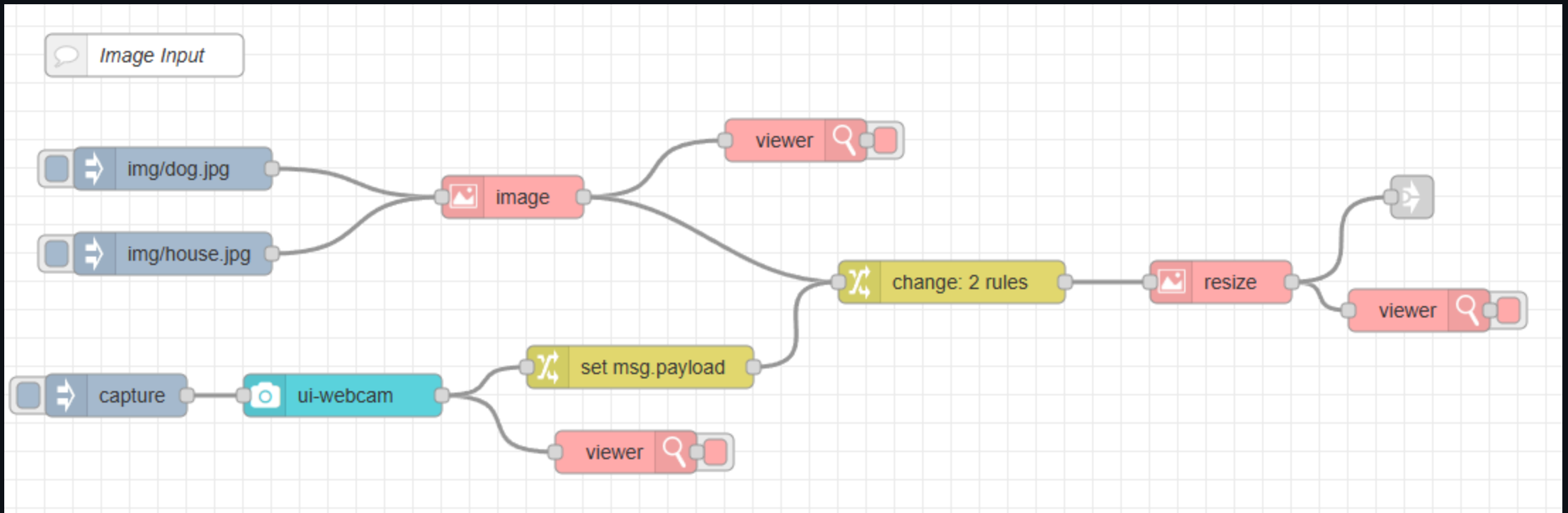
Install Additional Modules

- Go to Menu (top right) -> Manage palette -> Install
- Install the following modules:
 - `node-red-contrib-image-tools`
 - `@sumit_shinde_84/node-red-dashboard-2-ui-webcam`

Prepare Mock Images

- Copy image from `ml-express/temp/img` to your node-red installation directory under `img` folder.

Image input flow



Inject Node Configuration

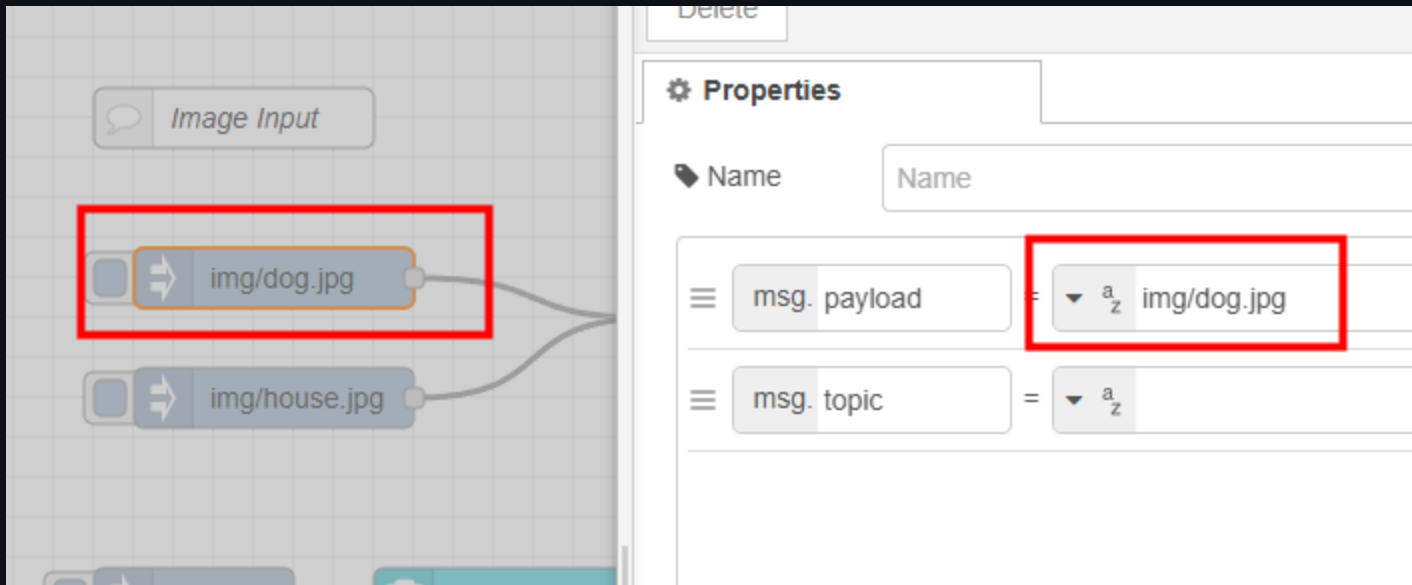
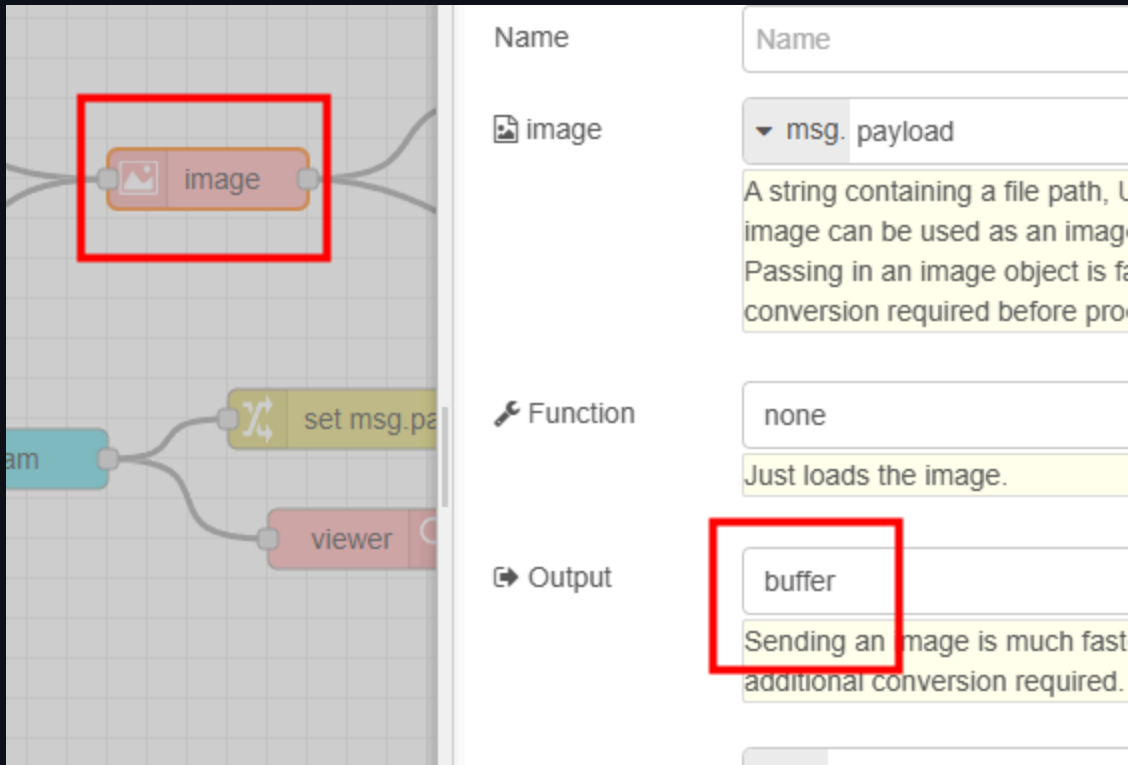
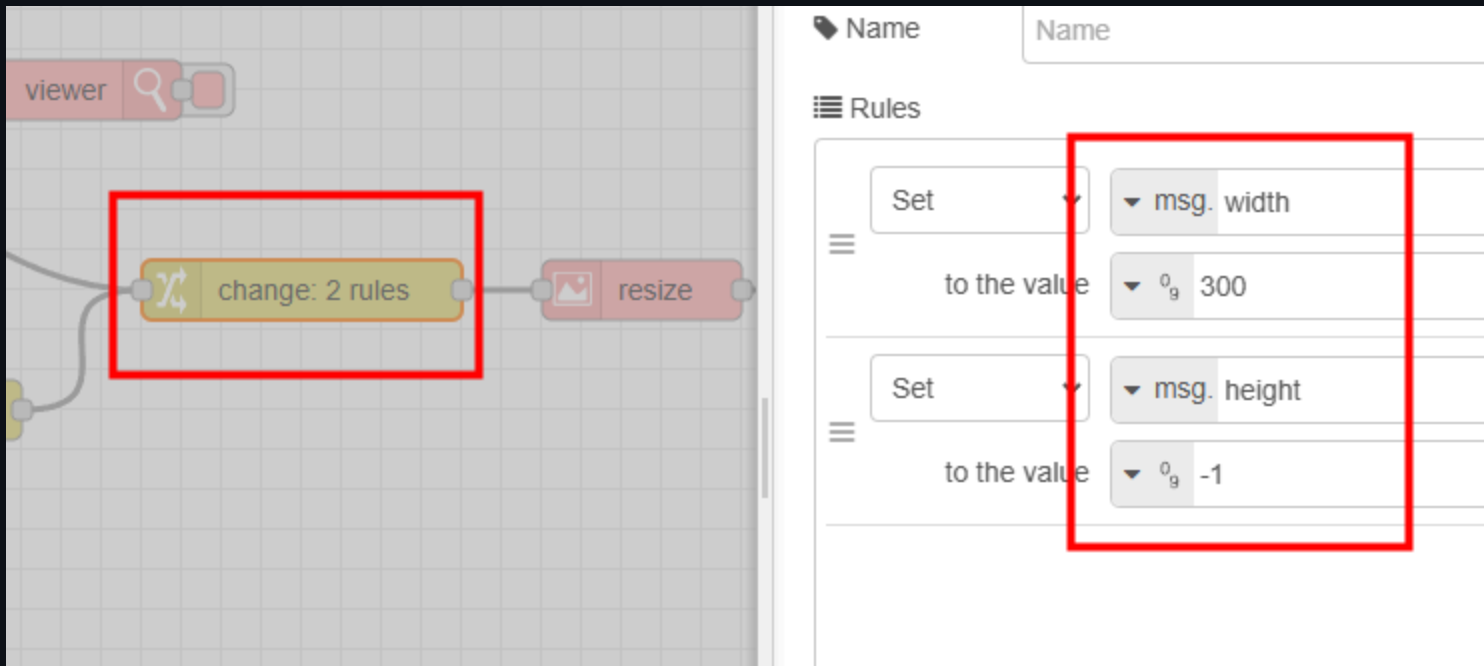


Image Node Configuration



Change output to **buffer**.

Change Node Configuration



Setting width and height options.

Image Node Configuration (Resize)

A string containing a file path, URL or base64 encoded image can be used as an image source. Note: Passing in an image object is faster as there is no conversion required before processing.

Function `resize`

Output `buffer`

Output Property `msg.payload`

</> w `msg.width`

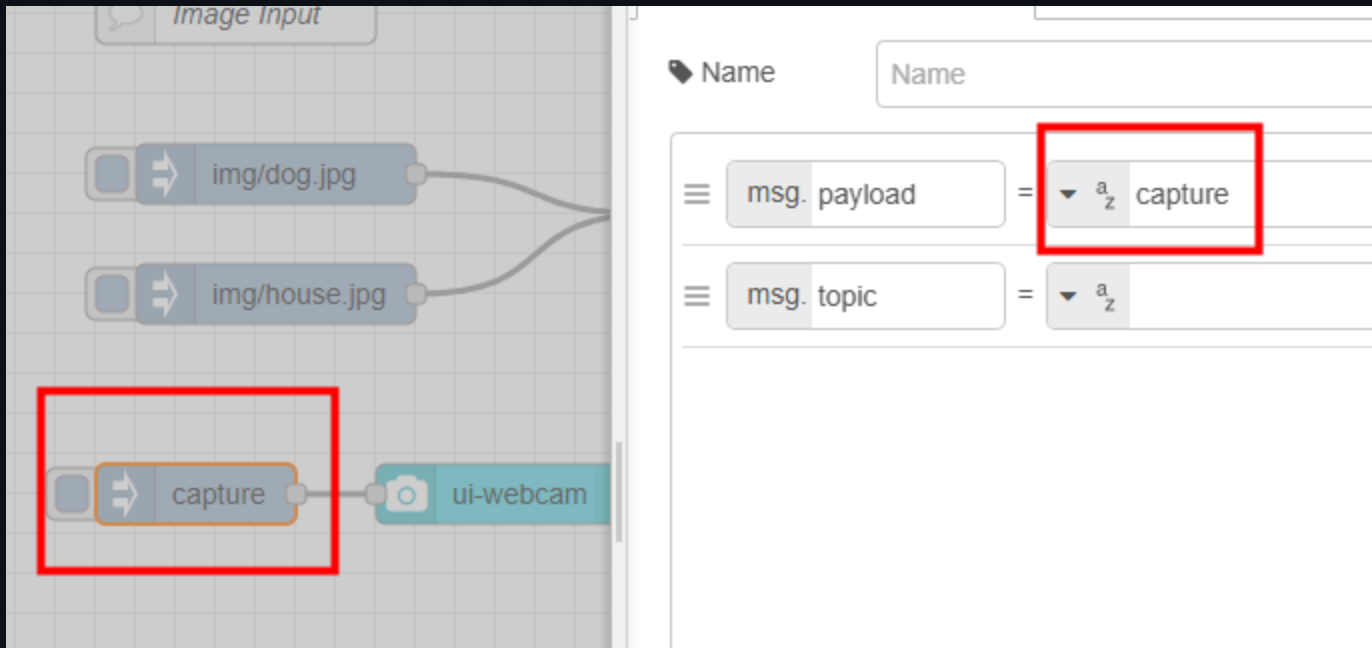
</> h `msg.height`

The msg property in which to send the resized image.

(Required) the width to resize the image to. Can be "Jimp.AUTO" or -1)

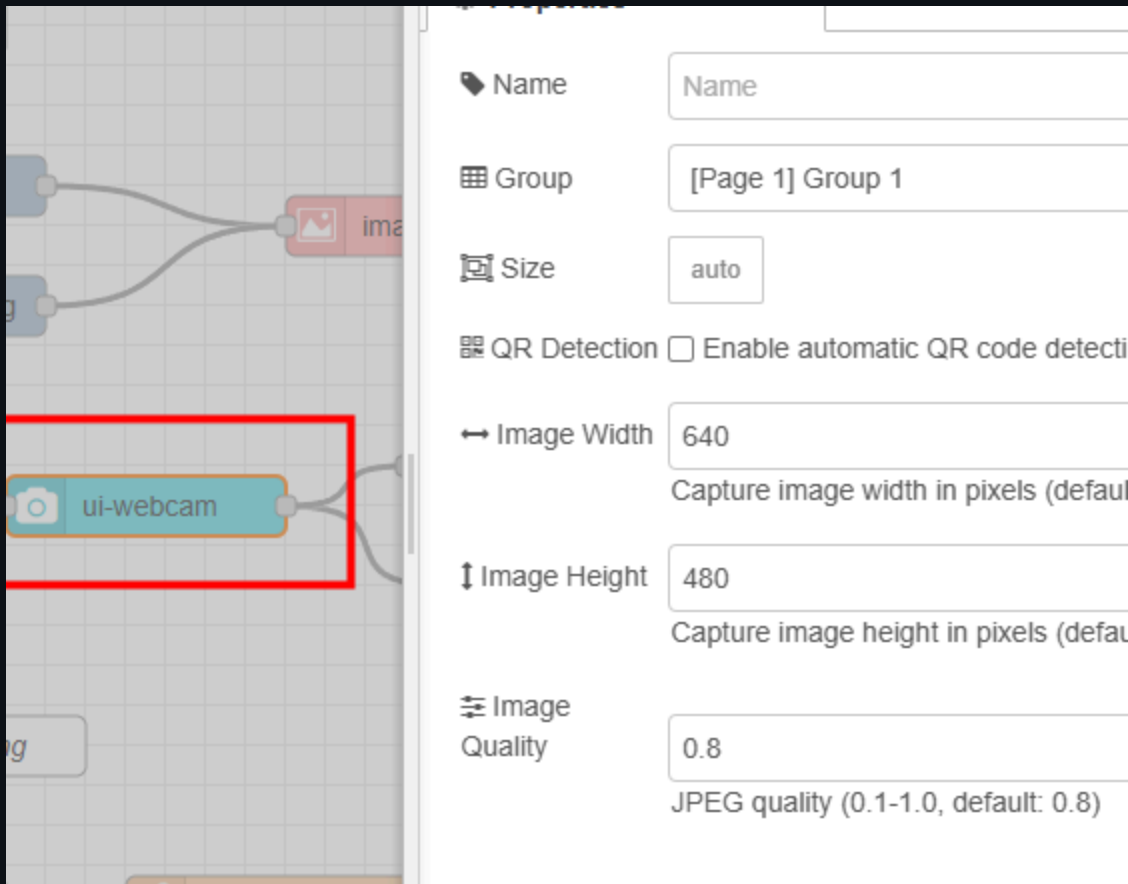
(Required) the height to resize the image to. Can be "Jimp.AUTO" or -1)

Inject Node Configuration (Capture Image)



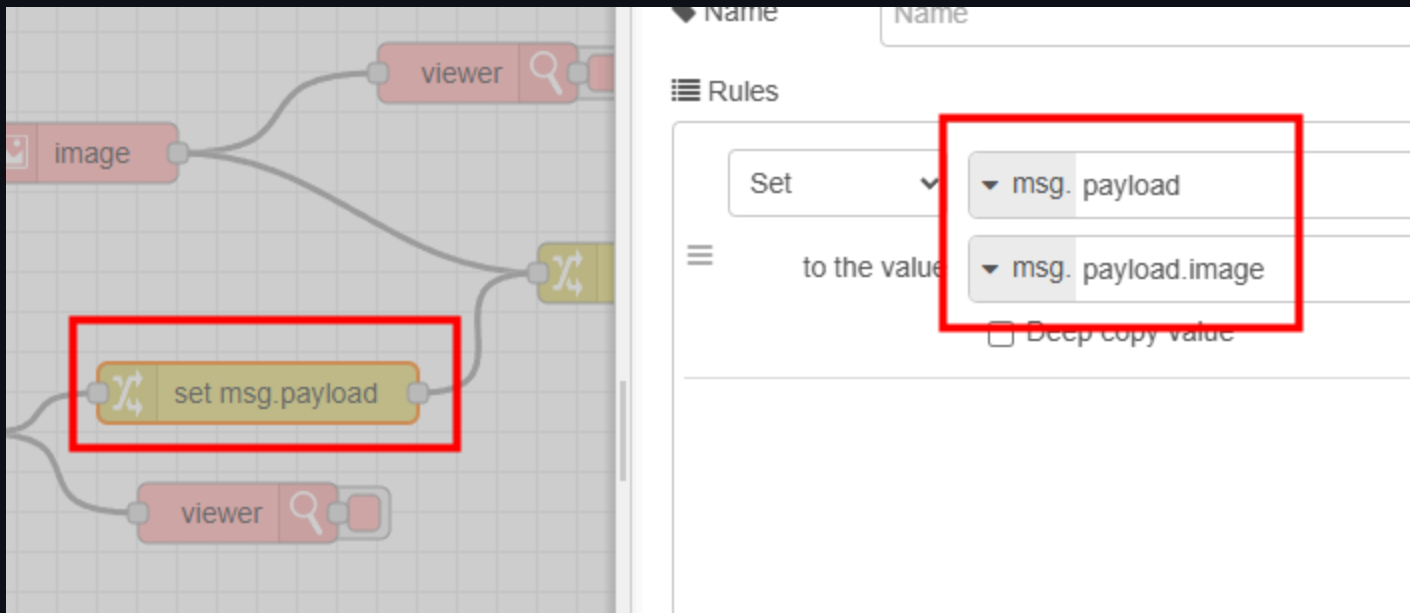
Set payload to `capture` to trigger webcam capture.

UI-Webcam Node Configuration



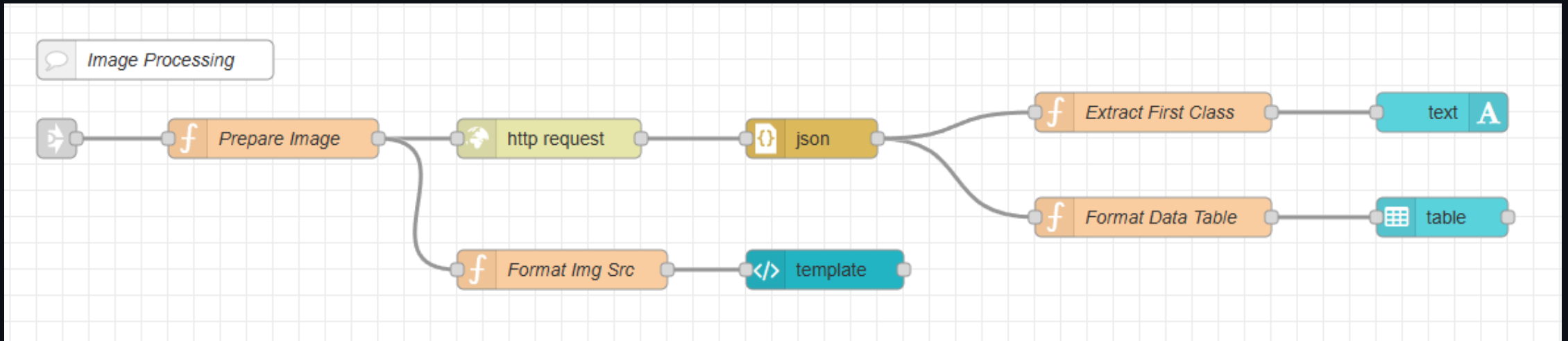
Leave default settings.

Change Node Configuration



Extract `image` field from webcam output.

Image-Processing Flow



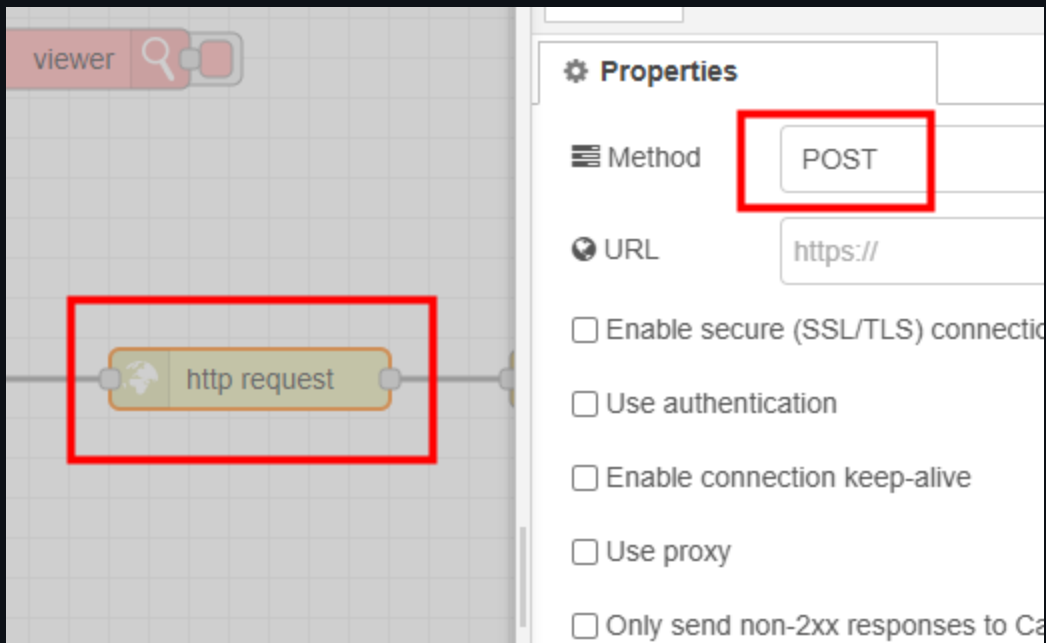
Note

- I attempted to use `complete` node to link payload from `image` node (resize), but it did not get triggered as expected.
 - This might be due to the way `image` node works internally. (Probably not calling `node.done()` function)
- I will use `link` nodes to connect the two flows instead.

Prepare Image Function Node

```
msg.payload = { imageEncoded: Buffer.from(msg.payload).toString("base64") };  
msg.headers = { "Content-Type": "application/x-www-form-urlencoded" };  
msg.url = "http://localhost:3003/upload_base64"; // ML server endpoint  
return msg;
```


HTTP Request Node Configuration



Extract First Class Function Node

```
const predictions = msg.payload?.predictions;
if (Array.isArray(predictions)) {
  msg.payload = predictions[0]?.class ?? "No Class Name";
} else {
  msg.payload = "No Class";
}
return msg;
```

Format Data Table Function Node

```
const predictions = msg.payload?.predictions;
if (!Array.isArray(predictions)) {
  msg.payload = [];
  return msg;
}

const dt = new Date();
const datestring = dt.toLocaleDateString();
const timestring = dt.toLocaleTimeString();
msg.payload = predictions.map((d) => ({
  Class: d["class"],
  Score: `${Math.round(d["score"] * 100)}%`,
  Time: `📅 ${datestring} ⌚ ${timestring}`,
})));
return msg;
```

Format Img Src Function Node

```
msg.payload = "data:image/jpeg;base64," + msg.payload.imageEncoded;  
return msg;
```

UI-Template Node Configuration

```
<template>
  <div>
    
  </div>
</template>

<script>
export default {};
</script>
<style></style>
```

Display image from base64 string.

The image shows a web development tool interface. On the left, a canvas displays a 'json' widget (yellow) and a 'template' widget (teal) connected by lines. The 'template' widget is highlighted with a red rectangle. On the right, the configuration panel for the 'template' widget is shown. It includes fields for 'Type' (Widget (Group-Scoped)), 'Group' ([Page 1] Group 1), 'Size' (auto), and 'Class' (Optional CSS class name(s)). Below these is a 'Template' section with a red-bordered text area containing the following code:

```
1 <template>
2   <div>
3     
4   </div>
5 </template>
6
7 <script>
8   export default {};
9 </script>
10 <style></style>
```

Train your own model

- Teachable machine
- Create an image classification model.
- Download model and extract to `ml-express/public/model` folder.